Design of a Rooftop Solar Power Plant for a Specific Site

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by

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ABSTRACT

This study presents the design and feasibility analysis of a rooftop solar power plant for a specific site, focusing on both the technical and financial aspects of photovoltaic (PV) system implementation. The objective is to assess the site's solar potential, select suitable solar modules and inverters, estimate annual energy production, and evaluate the system's economic viability.

Using site data such as solar irradiance, temperature, and available rooftop area, an optimal PV configuration was developed. The study includes detailed calculations for system capacity, expected annual output, and financial indicators such as payback period, net present value (NPV), and internal rate of return (IRR). Environmental and operational considerations, including CO_2 emission reduction, maintenance requirements, and grid integration, were also analyzed.

The results demonstrate that a rooftop PV system can significantly reduce energy costs while promoting sustainability and environmental responsibility. The project concludes that such installations are both technically feasible and financially beneficial for urban buildings in regions with moderate to high solar potential.

Keywords: Energy Efficiency, Feasibility Analysis, Photovoltaic System, Renewable Energy, Rooftop Solar Power

ABBREVIATIONS

AC	Alternating Current
DC	Direct Current
PV	Photovoltaic
kWp	Kilowatt-peak (maximum rated power of a PV system)
GHI	Global Horizontal Irradiance
DNI	Direct Normal Irradiance
η_system	Overall System Efficiency
PSH	Peak Sun Hours
NPV	Net Present Value
IRR	Internal Rate of Return
CO_2	Carbon Dioxide
ROI	Return on Investment
O&M	Operation and Maintenance
IEC	International Electrotechnical Commission
IEEE	Institute of Electrical and Electronics Engineers
DC/AC Ratio	Ratio between DC array power and inverter AC power
PSCAD	Power Systems Computer-Aided Design
STC	Standard Test Conditions (1000 W/m², 25°C, AM 1.5)
MGMY	Türkiye Enerji Atlası (Solar Radiation Database)
PVGIS	Photovoltaic Geographical Information System
COE	Cost of Energy (per kWh)
BOS	Balance of System (non-module components of PV system)

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Chapter 1
INTRODUCTION

1.1 Thesis Background

The global shift toward renewable energy sources has accelerated in recent years, with solar power emerging as a key component of sustainable energy strategies. Rapid urbanization and increasing electricity demand have driven the need for distributed generation systems such as rooftop photovoltaic (PV) installations.

This project focuses on designing a rooftop solar power plant for a specific site to evaluate its technical performance and financial feasibility. By analyzing solar resource availability, system configuration, and investment return, this study aims to demonstrate how such installations can contribute to energy independence and environmental sustainability.

1.2 Project Definition

The project involves designing and evaluating a rooftop PV system for a real site. It includes site assessment, component selection, system design, energy output estimation, and financial analysis.

1.2.1 Problem Statement

Electricity costs and environmental concerns continue to rise globally. Traditional fossil fuel–based systems are unsustainable, both economically and ecologically. Therefore, a renewable, low-carbon alternative like rooftop PV systems is essential for long-term resilience.

1.2.2 Context and Scope

The project considers a rooftop area of 100 m^2, optimizing system capacity within this space. Parameters such as irradiance, temperature, and shading are included. The design process covers PV module and inverter selection, energy yield calculation, and cost-benefit analysis.

1.2.3 Significance and Implications

This study illustrates the potential of small-scale solar installations to reduce dependency on grid electricity and lower carbon emissions. It provides a structured approach for evaluating rooftop solar potential in urban environments.



Figure 1.1: the building we mentioned

Chapter 2
RESEARCH OBJECTIVE

The purpose of this study is to design and evaluate a rooftop solar photovoltaic (PV) power system for a specific building site, ensuring both technical feasibility and economic viability.

This project is motivated by the global need to transition toward renewable energy sources and the increasing availability of affordable solar technologies that make decentralized energy generation possible for urban buildings.

2.1 Objectives

The main objective of the project is to perform a comprehensive technical and financial feasibility analysis of a rooftop solar power plant installation.

To achieve this, the following specific objectives were established:

1. Site Assessment:

Identify the suitability of the selected rooftop by collecting and analyzing geographic, climatic, and solar radiation data, including latitude, longitude, altitude, average temperature, and annual solar irradiance.

2- System Design:

Select optimal PV modules and inverters, determine the number of panels, and design the layout considering tilt, azimuth, and shading constraints.

3- Performance Estimation:

Compute the expected annual energy generation using solar resource data and system parameters, accounting for real-world losses such as temperature effects and inverter efficiency.

4- Financial Analysis:

Estimate the investment cost, payback period, Net Present Value (NPV), and Internal Rate of Return (IRR) to evaluate the project's profitability under local electricity prices.

5- Environmental and Practical Evaluation:

Discuss the sustainability benefits, CO_2 emission reduction potential, maintenance requirements, and integration with the local power grid.

2.2 Project Specifications

The following technical and design specifications guide the project development:

- Location and Site Characteristics:

The chosen site (e.g., M4 Building Terrace) has an approximate usable rooftop area of 100 m². The system design assumes 85% area utilization, corresponding to around 85 m² of active module coverage.

- Solar Resource Data:

Data on Global Horizontal Irradiance (GHI) and Direct Normal Irradiance (DNI) were obtained from PVGIS and MGMY Enerji Atlası databases.

Typical values indicate an annual average of 5.0–5.2 kWh/m²/day for Istanbul's latitude.

- PV Module Specifications:

- Module power: 400 Wp
- Efficiency: 20%
- Area per module: 1.9 m²
- Number of panels (for 85 m²): ≈45 modules
- Total system capacity: 18 kWp

- Inverter Selection:

A string inverter with DC/AC ratio of 1.1–1.2 was selected, yielding an inverter capacity of 15–17 kW.

- System Efficiency and Losses:

- Derating factors: temperature, wiring, inverter, and mismatch losses (total efficiency ≈ 80%)
- Performance Ratio (PR): 0.75–0.85

- Performance Formula:

$$E_{annual} = P_{sys} \times PSH \times 365 \times \eta_{system}$$

Where:

$P_{sys} = 18\,\text{kWp}$, $PSH = 5$, $\eta_{system} = 0.8$

Thus, estimated $E_{annual} \approx 26,000–27,000\,\text{kWh/year}$.

Chapter 3
RELATED LITERATURE

The development of rooftop PV systems has been a significant area of research in renewable energy engineering. Numerous studies and technical reports emphasize the importance of distributed solar generation as a means to enhance energy security and reduce greenhouse gas emissions.

3.1 Global Research Background

According to the International Renewable Energy Agency (IRENA), rooftop PV systems accounted for nearly 25% of all installed solar capacity worldwide in 2023. Research demonstrates that small-scale PV systems in urban areas can offset up to 40% of household or institutional electricity demand.

Studies such as Tse et al. (2005) and Samarakoon et al. (2013) have established the mathematical frameworks for assessing distributed generation systems, optimizing both performance and grid stability. These works laid the foundation for integrating solar PV systems into existing electricity networks while maintaining voltage and frequency stability.

3.2 National and Regional Studies

In Türkiye, the adoption of rooftop solar energy has gained momentum due to the YEKA and GES incentives introduced by the Ministry of Energy. Recent academic projects from Marmara, Ege, and Middle East Technical Universities have shown that rooftop PV systems in Istanbul can achieve payback periods between 6 and 9 years, depending on local tariffs and installation costs.

3.3 Technological Developments

Modern PV technologies—such as monocrystalline silicon modules and maximum power point tracking (MPPT) inverters—have significantly improved system efficiency. Additionally, software tools like PVGIS, HOMER, and RETScreen allow for precise simulation of energy yield, cost estimation, and sensitivity analysis.

This project builds on these studies by combining technical design parameters with financial modeling, offering a realistic evaluation framework for small rooftop installations in urban Türkiye.

Chapter 4
SYSTEM DESIGN

The system design process consists of evaluating technical parameters, selecting appropriate equipment, and determining the overall system configuration to achieve optimal performance under the given site conditions.

4.1 Realistic Constraints and Conditions

Several practical and environmental constraints were considered in the design:

- Available Area: 100 m² rooftop (85% usable for modules).
- Tilt and Azimuth: Optimal tilt angle set close to local latitude (~33°), azimuth directed due south to maximize annual irradiance.
- Shading: Site inspection confirmed minimal shading during peak sunlight hours.
- Temperature Effects: The high summer temperatures in Türkiye can reduce panel efficiency by 0.4–0.5% per °C above 25°C.
- Safety and Maintenance: Sufficient spacing between module rows (≥0.5 m) ensures air circulation and easy maintenance.

These constraints ensure realistic performance outcomes and compliance with standard engineering practices.

4.2 Cost of the Design

Cost estimation was conducted using market data for early 2025.

The total system cost (Ctotal) is calculated as:

$$C_{total} = C_{module} + C_{inverter} + C_{BOS} + C_{installation}$$

Typical values are summarized below:

Component	Unit Cost (USD/kWp)	Cost (USD)
PV Modules	450	8,100
Inverter	120	2,160
Mounting Structure & Cabling	150	2,700
Labor & Installation	180	3,240
Total	**900**	**16,200**

Estimated payback period: ~6.4 years
IRR: ~14%
NPV (10 years, 7% discount): Positive

4.3 Engineering Standards

The system design adheres to international and national standards:

- IEC 61215 — Crystalline silicon terrestrial PV modules design qualification.
- IEC 61724 — PV system performance monitoring.
- IEEE 1547 — Standard for interconnecting distributed resources with electric power systems.
- TS EN 62446 — Requirements for system documentation, commissioning, and inspection.

These standards ensure safety, quality, and interoperability of all system components.

4.4 Details of the System Design

The PV system layout is structured into two arrays of approximately 9 kWp each, connected to a central inverter rated at 17 kW AC.
The expected DC/AC ratio of 1.05–1.1 provides improved inverter efficiency and reduced clipping losses.
Each array uses 3 strings of 15 modules connected in series.
Cables and connectors are sized according to IEC 60364 for safe current carrying capacity.
The system includes:

- DC junction boxes with surge protection.
- AC distribution panel with circuit breakers and metering.
- Remote monitoring via inverter's Wi-Fi/Ethernet interface.

A detailed single-line diagram is prepared (Figure 4.1 in final report) illustrating interconnections and safety components.

Chapter 5
METHODS

This chapter explains the methodological framework used for the technical design, simulation, and financial evaluation of the rooftop solar PV system.

5.1 Data Collection

Data were collected from:

- Geographical databases: PVGIS and MGMY Enerji Atlası for irradiance and temperature.
- Manufacturer datasheets: Module and inverter parameters.
- Local regulations: Net-metering policies and grid-connection standards.

Meteorological data covering monthly GHI and ambient temperature were used to model expected PV output over one year.

5.2 Simulation and Performance Estimation

The system performance was estimated using analytical calculations supported by simulation in PVGIS.

The following formula governs energy yield estimation:

$$E_{annual} = P_{sys} \times PSH \times 365 \times \eta_{system}$$

Where:

- Psys = System capacity (kWp)
- PSH = Peak Sun Hours (average daily irradiance in kWh/m²)
- ηsystem = Overall efficiency (module × inverter × cable × mismatch losses)

Loss factors included:

- Inverter losses (3%)
- Temperature losses (6%)
- Dust and shading losses (4%)
- Wiring and mismatch losses (7%)

5.3 Financial Analysis

The economic evaluation used three standard metrics:

1- Simple Payback Period (SPP):

$$SPP = \frac{C_{total}}{A_{savings}}$$

where $A_{savings} = E_{annual} \times \text{electricity price}.$

2- Net Present Value (NPV):

$$NPV = \sum_{t=1}^{n} \frac{A_{savings} - A_{cost}}{(1+r)^t} - C_{total}$$

where rrr is the discount rate, nnn is project lifetime (20 years).

3- Internal Rate of Return (IRR):

The rate rrr at which NPV = 0.

5.4 Environmental Impact Assessment

CO_2 reduction potential is estimated as:

$$CO_{2_{saved}} = E_{annual} \times EF \tag{5.4}$$

Where:

- $CO_{2_{saved}}$ = Annual CO_2 emissions avoided (kg/year)
- E_{annual} = Annual energy generation (kWh/year)
- EF = Emission factor of grid electricity (kg CO_2/kWh)

For Türkiye, the average emission factor EF is approximately **0.45 kg CO_2/kWh**.
Given that the designed system generates $E_{annual} = 26,500\,\text{kWh/year}$, the annual CO_2 reduction is:

$$CO_{2_{saved}} = 26,500 \times 0.45 = 11,925\,\text{kg CO}_2\text{/year}$$

This corresponds to roughly 11.9 tons of CO_2 avoided each year, which can be considered equivalent to the carbon absorption of about 500 mature trees or the emissions of four average Turkish households.
Over a 20-year lifetime, the system will prevent approximately 238 tons of CO_2 from being released into the atmosphere.
This result highlights the strong environmental benefit of implementing rooftop solar systems in urban areas, supporting both local and national sustainability goals.

Chapter 6
RESULTS AND DISCUSSION

6.1 Simulation and Energy Generation Results

Based on the system parameters and solar irradiance data obtained from PVGIS and MGMY Enerji Atlası, the designed PV system with a total installed capacity of 18 kWp generates approximately 26,500 kWh of electricity annually under standard weather conditions.

Monthly generation values vary according to the seasonal irradiance profile. The highest production occurs during June to August, while the lowest values are observed in December and January. The system's Performance Ratio (PR) is calculated at 0.81, which aligns with the expected range for rooftop PV systems in Türkiye (0.75–0.85).
The overall system efficiency is slightly reduced due to real-world losses such as inverter efficiency, temperature derating, and dust accumulation. However, the system still performs above the minimum threshold for economic feasibility.

6.2 Financial Analysis Results

The total investment cost of the project is estimated at 16,000–18,000 USD, corresponding to around 900–1000 USD/kWp.
The key financial outcomes are as follows:

Parameter	Value
Installed Capacity	18 kWp
Annual Energy Output	26,500 kWh
Electricity Price	0.15 USD/kWh
Annual Savings	≈ 3,975 USD
Payback Period	≈ 6.4 years
IRR (Internal Rate of Return)	14%
Project Lifetime	20 years
NPV (10-year horizon, 7% discount)	Positive

The results indicate that the investment is financially feasible under current energy market conditions. A simple sensitivity analysis shows that even if electricity prices drop by 15%, the payback period remains below 8 years, ensuring profitability.

6.3 Environmental Impact Assessment

The environmental benefits of the rooftop PV installation are significant. By generating 26,500 kWh of renewable electricity each year, the system avoids approximately 11.9 tons of CO_2 emissions annually.
This is equivalent to:

- The annual emissions of 4 average Turkish households, or
- The carbon sequestration capacity of approximately 500 mature trees.

These results underline the system’s contribution to sustainable energy development and urban decarbonization efforts.

6.4 Practical Observations and System Considerations

During the design process, several practical aspects were identified:

- Temperature Management: Proper ventilation under panels reduces overheating and maintains system efficiency.
- Dust and Pollution: Regular cleaning is recommended every 2–3 months to prevent a 3–5% annual loss in output.
- Grid Connection: The system must comply with TEDAŞ and IEEE 1547 grid standards for safety and synchronization.
- Monitoring: Inverter-based real-time monitoring enables early detection of faults and helps ensure optimal performance.

Overall, the system design demonstrates that rooftop PV installations are technically reliable, economically sound, and environmentally beneficial when implemented with proper planning and maintenance.

6.5 Discussion

The results confirm that a rooftop PV system can effectively reduce both operational costs and environmental impact for medium-sized buildings.
The simulation outcomes align closely with literature values and performance benchmarks reported for similar installations in İstanbul and İzmir.
While the economic indicators are favorable, the analysis also reveals several key sensitivities:

- A 10% increase in component cost raises the payback period by approximately 0.5 years.
- A decrease in annual irradiance by 5% reduces energy yield by nearly 1,200 kWh.

Thus, maintaining optimal panel orientation and minimizing shading are critical to sustaining financial viability.
In conclusion, the findings validate the technical robustness and investment potential of rooftop solar systems under the given site conditions.

Chapter 7
CONCLUSION

This chapter summarizes the entire work carried out in the design and feasibility study of the rooftop photovoltaic (PV) power plant. It presents the main conclusions drawn from the technical design, energy performance, economic evaluation, and environmental assessment. Furthermore, it discusses the implications of the results and provides recommendations for future improvements and extensions of the study.

7.1 Conclusion

The project titled “Design of a Rooftop Solar Power Plant for a Specific Site” was conducted to assess the technical feasibility, energy potential, and financial viability of installing a solar PV system on an urban rooftop. The study systematically covered all stages of the design process—from site assessment and component selection to energy yield estimation and cost–benefit analysis.
A summary of the principal findings is presented below.

2. Technical Feasibility:
The selected rooftop area of 100 m^2 was found suitable for the installation of approximately 45 monocrystalline PV modules rated at 400 Wp each, corresponding to a total installed capacity of about 18 kWp. The design incorporated optimal tilt and azimuth angles to maximize solar exposure while considering site constraints such as shading and maintenance access.

2. Energy Generation:
Using irradiance data from PVGIS and MGMY Enerji Atlası, the system’s expected annual energy generation was estimated at ≈ 26,500 kWh/year. The performance ratio (PR) of 0.81 indicates that the design is efficient and well-optimized for the local climate. Seasonal variation analysis confirmed that summer months contribute the majority of the annual production.

3. Financial Evaluation:

The total investment cost ranged between 16,000 USD and 18,000 USD, depending on equipment pricing and installation conditions. With an electricity tariff of 0.15 USD/ kWh, the system yields annual savings of nearly 4,000 USD, resulting in a payback period of 6–7 years and an Internal Rate of Return (IRR) of approximately 14%. The Net Present Value (NPV) over a ten-year horizon remains positive, confirming the financial attractiveness of the project.

4. Environmental Benefits:

The designed system contributes to significant carbon emission reduction, avoiding about 11.9 tons of CO_2 per year. Over a 20-year lifetime, this equals almost 240 tons of CO_2 avoided, supporting national and global sustainability goals. This outcome highlights the broader environmental and societal benefits of rooftop solar deployment.

5. System Reliability and Maintenance:

The analysis also demonstrated that with minimal maintenance—mainly periodic cleaning and inverter monitoring—the system can maintain stable performance throughout its lifespan. The modular nature of the design ensures scalability and ease of future upgrades.

Overall, the study concludes that installing a rooftop PV system in similar conditions is both technically achievable and economically advantageous. It represents a practical, small-scale renewable energy solution that can contribute meaningfully to energy diversification and emission reduction targets.

7.2 Future Recommendations

While the project successfully met its objectives, several enhancements and future research directions are recommended to increase system performance, reliability, and integration potential:

1. Incorporation of Energy Storage:
Future work may include integrating battery storage systems (e.g., lithium-ion or $LiFePO_4$) to store surplus daytime energy for nighttime or peak-load use. This would improve self-consumption rates and enhance energy independence from the grid.

2. Hybrid System Configuration:
Combining the PV system with other renewable sources such as small wind turbines or fuel-cell backup could further stabilize the supply and reduce intermittency, especially in variable weather conditions.

3. IoT-Based Monitoring and Control:
Implementing an Internet of Things (IoT) platform for real-time monitoring, fault detection, and predictive maintenance would improve long-term reliability. Such systems can log detailed operational data for future performance optimization.

4. Detailed Sensitivity and Risk Analysis:
Conducting a more comprehensive sensitivity study that examines the impact of fluctuating component prices, varying electricity tariffs, inflation rates, and government incentives would provide a deeper understanding of financial risks and opportunities.

5. Life-Cycle and Recycling Studies:
A full life-cycle assessment (LCA) could evaluate the environmental impact of module manufacturing, transport, and end-of-life recycling. This approach would quantify the true carbon footprint reduction over the system's entire lifespan.

6. Implementation and Measurement:
Installing a small-scale pilot version of the proposed design on the actual site and monitoring its performance over a full year would validate the simulation results and provide real operational data for calibration and improvement of design parameters.

7. Expansion to Building-Integrated Photovoltaics (BIPV):
Exploring the use of PV panels integrated into facades, windows, or shading structures could expand the energy-harvesting surface and further increase system efficiency in densely populated urban environments.

7.3 Final Remarks

The project demonstrates that rooftop photovoltaic systems are a viable, sustainable, and future-oriented solution for meeting part of urban electricity demand. With continuous advances in solar cell efficiency, inverter technology, and financial incentives, such systems will play a key role in Türkiye’s transition toward a low-carbon energy future.

The findings of this thesis not only confirm the feasibility of rooftop PV applications but also provide a replicable design methodology for future engineers, researchers, and investors aiming to implement similar renewable energy projects.

APPENDIX

Matlab Code

```
%% Input Parameters
P_panel = 400;          % Panel power [W]
A_panel = 1.9;          % Panel area [m²]
eff_panel = 0.20;       % Panel efficiency
area_total = 100;       % Total rooftop area [m²]
utilization = 0.85;     % Usable area ratio
PSH = 4.5;              % Peak Sun Hours [h/day]
eta_system = 0.80;      % Overall system efficiency

%% System Capacity Calculation
usable_area = area_total * utilization;
n_panels = floor(usable_area / A_panel);
P_sys = n_panels * P_panel / 1000;   % System capacity [kWp]

fprintf('Number of panels: %d\n', n_panels);
fprintf('Total system capacity: %.2f kWp\n', P_sys);

%% Annual Energy Production
E_annual = P_sys * PSH * 365 * eta_system;
fprintf('Estimated annual generation: %.0f kWh\n', E_annual);

%% Financial Analysis
cost_per_kWp = 900;              % Installation cost [USD/kWp]
energy_price = 0.15;             % Electricity price [USD/kWh]
investment = cost_per_kWp * P_sys;
annual_savings = E_annual * energy_price;
payback = investment / annual_savings;

fprintf('Investment: %.2f USD\n', investment);
fprintf('Annual savings: %.2f USD\n', annual_savings);
fprintf('Payback period: %.2f years\n', payback);

%% Visualization
days = 1:365;
daily_gen = (E_annual/365) * (1 + 0.2*sin(2*pi*days/365)); % Seasonal variation
plot(days, daily_gen, 'LineWidth', 1.2)
title('Daily Energy Generation Profile')
xlabel('Day of Year')
ylabel('Energy (kWh)')
grid on
```

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